Introduction

Almost every journal-worthy figure is a composition: two or more panels arranged with shared legends, alphabetical tags, a unifying title, and aligned axes. Building such figures by hand with `gridExtra` or `cowplot` is verbose and brittle to layout changes. The `patchwork` package replaces that workflow with arithmetic-like operators that mirror the way the figure is read: `+` for "next to", `/` for "below", `|` for "in a row". Combined with `plot_layout()` for relative sizes and `plot_annotation()` for shared titles and tags, it is now the standard tool for assembling multi-panel `ggplot2` figures.

Prerequisites

A working knowledge of `ggplot2` — geoms, scales, themes — and an idea of which combinations of plots actually communicate (rather than merely fill space).

Theory

`patchwork` treats each `ggplot` as an object that supports a small algebra:

- `p1 + p2` places two plots side by side at equal width.
- `p1 / p2` stacks them vertically.
- Parentheses control nesting: `(p1 | p2) / p3` puts `p1` and `p2` horizontally with `p3` spanning below.
- `plot_layout(guides = "collect")` deduplicates shared legends, moving a single copy to the figure margin.
- `plot_annotation(title = ..., tag_levels = "A")` adds a global title and panel tags (A, B, C, ...) automatically.
- `& theme_minimal()` applies a theme to every panel in the composition simultaneously, useful when individual panels were built without a consistent theme.

The package handles axis alignment automatically when plots share axis ranges, which avoids the tiny horizontal offsets that bedevil hand-aligned multi-panel figures.

Assumptions

No statistical assumptions; the only practical constraint is that combined panels must remain legible at the final printed size, which usually means three to four panels per row at most.

R Implementation

```r
library(ggplot2); library(patchwork)

p1 <- ggplot(mtcars, aes(wt, mpg))       + geom_point() + theme_minimal()
p2 <- ggplot(mtcars, aes(factor(cyl)))   + geom_bar()   + theme_minimal()
p3 <- ggplot(mtcars, aes(hp))             + geom_histogram(bins = 15) + theme_minimal()

# Two side by side
p1 + p2

# Stacked
p1 / p2

# Complex layout
(p1 | p2) / p3 +
  plot_annotation(title = "mtcars overview", tag_levels = "A")
```

```r
# Collect shared legends
p1 + p2 + plot_layout(guides = "collect")
```

Output & Results

Four progressively more complex layouts: two horizontal panels, two vertical panels, a 2-up-1-down arrangement with global title and panel tags A, B, C, and a side-by-side composition with one shared legend collected into the right margin. Saving any of these with `ggsave()` produces a single file at the specified dimensions, ready for journal submission.

Interpretation

A reporting sentence (figure caption): "Figure 2 (assembled with `patchwork`): A) weight vs miles-per-gallon, B) cylinder distribution, C) horsepower distribution; panels share a common minimal theme and a single legend." Caption-level credit to the composition tool is unnecessary, but tags and a one-line description of each panel are essential for any multi-panel figure.

Practical Tips

- Use `plot_layout(widths = c(2, 1))` to give one panel twice the horizontal space of another; `heights` does the same vertically.
- `tag_levels = "A"` produces A, B, C tags; `"1"` produces 1, 2, 3; `"i"` produces lowercase Roman numerals.
- `& theme_minimal()` (note the `&`) applies a theme globally to every panel; `+` only applies to the most recently added plot.
- For many small panels with the same structure, prefer `facet_wrap()` inside a single `ggplot` rather than composing dozens of plots; faceting is more efficient and produces aligned axes by default.
- Always save the composed plot as a single file via `ggsave()`; assembling separate exports in a vector editor produces fragile artwork.
- When a shared legend has many entries, place it on the right via `plot_layout(guides = "collect") & theme(legend.position = "right")` rather than letting it stack vertically inside one panel.

R Packages Used

`patchwork` for composition, `ggplot2` for the underlying plots, and `cowplot` for the alternative `plot_grid()` API and the `draw_image()` helper when embedding bitmap insets is required.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- ggplot2 grammar and multi-panel layouts